
MaxionBench-Portable: Decision-Centered Evaluation of Retrieval Infrastructure for Agentic AI

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Abstract

1 Modern agentic AI systems use retrieval infrastructure as operational memory:
2 they repeatedly read, write, and pack retrieved evidence into model contexts under
3 latency and cost constraints. Deployment choices, however, are often made from
4 quality-only leaderboards or from system benchmarks that omit agent-facing fresh-
5 ness and context-cost effects. We present MaxionBench-Portable, a conformance-
6 gated, single-node benchmark for comparing retrieval engines under three agentic
7 workloads: single-hop corpus retrieval, streaming memory with post-insert fresh-
8 ness probes, and multi-hop evidence retrieval. The benchmark evaluates four
9 reportable engines, two local embedding models, and three budget levels, then sep-
10 arates strict-latency deployment decisions from unconstrained-cost and quality-first
11 choices. On the archived portable run, a 200 ms worst-case p99 constraint selects
12 FAISS CPU with bge-small for all three workloads, but the decision margins differ:
13 S1 and S2 are cost/quality ties resolved by latency, while S3 has lower cost and
14 higher evidence coverage than the next strict-cost alternative. Paired audits rule
15 out two tempting overclaims: Qdrant does not hide an S2 quality/freshness win
16 in the larger same-machine check, and the S3 pgvector quality-first margin is not
17 substantive under matched queries. The results also show why benchmark budgets
18 should be reported explicitly: B0-to-B2 top-1 decisions change for single-hop and
19 streaming workloads, whereas the multi-hop workload keeps the same top-1 rec-
20 ommendation despite low full-rank correlation. MaxionBench-Portable therefore
21 does not claim universal stability; it provides a reproducible protocol for deciding
22 when local benchmark evidence is stable enough for deployment.

1 Introduction

24 Retrieval infrastructure is becoming operational memory for agentic AI systems. Agents repeatedly
25 query, update, and reuse external stores while operating under latency budgets and context-window
26 costs, so a retrieval engine that looks strong on a static relevance benchmark may be unsuitable
27 when inserts must become visible quickly, tail latency must remain bounded, or retrieved passages
28 dominate downstream model cost. This paper studies the narrower question that an operator must
29 answer before deploying an agentic retrieval stack: can a commodity-hardware benchmark produce
30 stable deployment decisions for retrieval infrastructure?

31 Existing evidence is fragmented across two evaluation styles. Information-retrieval benchmarks such
32 as BEIR [Thakur et al., 2021] and HotpotQA [Yang et al., 2018] provide reusable relevance judgments,
33 but they do not specify engine conformance, update visibility, or tail-latency constraints. Systems
34 benchmarks often report throughput and latency, but they may not connect those measurements to
35 retrieval quality, freshness after writes, or the token cost induced by retrieved contexts. This gap
36 makes it easy to overfit a deployment decision to a single metric: a system can be fastest, cheapest, or
37 highest quality under different objectives.

MaxionBench-Portable addresses this gap with a reproducible, single-node benchmark for agentic retrieval infrastructure. The benchmark uses a fixed adapter contract, conformance tests before reporting, three workload scenarios, two local embedding tiers, and a budget ladder that measures whether decisions are stable as runs become longer. The central design choice is to treat benchmarking as a deployment decision audit rather than only a leaderboard: for each workload, MaxionBench-Portable identifies the minimum viable engine and embedding under a strict p99 latency threshold, separately reports unconstrained-cost and quality-first alternatives, and exposes the margins that make a decision robust or fragile.

The archived evaluation gives three concrete findings. First, under a 200 ms worst-case p99 threshold, FAISS CPU with bge-small is the minimum viable deployment for single-hop, streaming-memory, and multi-hop evidence retrieval. The decision-margin audit shows that this result should be read locally: S1 and S2 are cost/quality ties resolved by tail latency, whereas S3 has lower cost and higher evidence coverage than the next strict-cost candidate. Targeted paired audits check the most likely counter-stories: a larger same-machine S2 run finds no hidden Qdrant quality or freshness win, and a 5,000-query S3 audit demotes the pgvector quality-first row to objective-sensitivity evidence rather than a substantive quality advantage. Second, changing the objective can change the selected engine or embedding tier, so the benchmark should report strict-latency, unconstrained-cost, and quality-first decisions separately. Third, decision stability is workload-dependent: B0-to-B2 top-1 agreement is absent for S1 and S2 but present for S3, even though S3 has low full-rank Spearman correlation. These results support a cautious conclusion: commodity-hardware benchmarks can produce useful deployment evidence, but only when the paper reports conformance, budgets, decision margins, and objective sensitivity explicitly.

This paper contributes: (i) a conformance-gated benchmark protocol for agentic retrieval infrastructure; (ii) three portable workloads covering static retrieval, streaming memory, and bounded multi-hop evidence retrieval; (iii) a deployment-decision analysis that separates strict-latency, cost, quality, decision-margin, and paired-audit evidence; and (iv) an archived result bundle with confidence intervals, behavior-card support tables, metadata, and reproducibility commands.

2 Benchmark design

2.1 Scope and systems under test

MaxionBench-Portable evaluates retrieval infrastructure rather than full agent task success. The benchmark is intentionally single-node: all experiments run on one commodity operator machine, and the paper does not make distributed-systems or GPU-scaling claims. The reportable engine set contains FAISS CPU [Johnson et al., 2019], LanceDB in-process [LanceDB, 2024], PostgreSQL with pgvector [pgvector contributors, 2024], and Qdrant [Qdrant contributors, 2024]. LanceDB service mode is listed but excluded from the paper matrix because the archived conformance table has no passing local service row.

MaxionBench-Portable is organized as a decision-audit pipeline rather than as a single leaderboard: a reported winner is valid only under a stated conformance gate, workload, objective, and run budget.

Every reportable engine must pass the same adapter contract before its runs can enter paper-facing tables. The contract covers lifecycle operations, bulk and incremental writes, query and batch query, vector and payload updates, delete semantics, flush or commit behavior, index controls, and minimum stats fields. The conformance suite checks CRUD correctness, post-flush visibility, filter correctness, update semantics, delete semantics, and streaming-memory freshness semantics. This gate prevents a benchmark table from silently mixing engines with different unsupported behavior and is the main reason the paper treats missing conformance as an exclusion rather than a bad score.

2.2 Workloads

MaxionBench-Portable uses three workloads designed to isolate retrieval-infrastructure behavior that matters to agentic applications. S1 is single-hop retrieval over SciFact and FiQA through local BEIR-format bundles [Thakur et al., 2021, Wadden et al., 2020, Maia et al., 2018]; its primary quality metric is nDCG@10. S2 combines a deterministic 50K-passage static background from the same source family with a CRAG-500 event stream [Facebook Research, 2024]; it reports static nDCG@10 and freshness probes at one and five seconds after insert acknowledgment. S3 uses

Table 1: Portable benchmark main results. Quality and freshness report 95% confidence intervals; p99 reports the maximum observed tail latency across selected runs.

Workload	Engine	Emb.	Quality	Fresh@5	p99 max	Stability
S1 single-hop	faiss-cpu	bge-small	nDCG@10 0.506 (0.494–0.516)	–	11.2 ms	top-1 changed
S2 streaming	faiss-cpu	bge-small	nDCG@10 0.506 (0.487–0.524)	0.830 (0.805–0.852)	10.0 ms	top-1 changed
S3 multi-hop	faiss-cpu	bge-small	Cov@10 0.852 (0.849–0.854)	–	22.8 ms	top-1 stable

90 HOTPOTQA-PORTABLE, a frozen preprocessing of the HotpotQA dev distractor split, and measures
 91 whether the retrieved top-10 contains the supporting evidence set with evidence_coverage@10.

92 The workloads use fixed concurrency pins to keep the protocol small enough for a one-day local run.
 93 S1 and S3 evaluate read clients $\{1, 4, 8\}$. S2 uses eight read clients and two serialized write clients,
 94 then probes each inserted event passage at $T + 1s$ and $T + 5s$, where T is the time at which the
 95 adapter’s insert plus flush path returns. Each workload is evaluated with two local embedding tiers,
 96 BAAI/bge-small-en-v1.5 and BAAI/bge-base-en-v1.5 [Beijing Academy of Artificial Intelligence,
 97 2023].

98 2.3 Metrics and decisions

99 MaxionBench-Portable reports quality, tail latency, throughput, task-cost estimates, and stability
 100 across budget levels. For S1 and S2, quality is nDCG@10. For S3, quality is

$$\text{evidence_coverage@k}(q) = \frac{|E(q) \cap R_k(q)|}{|E(q)|},$$

101 where $E(q)$ is the gold evidence set and $R_k(q)$ is the retrieved top- k set. For S2, freshness is a binary
 102 hit if the inserted supporting passage appears in $R_{10}(q, T + \Delta)$ for $\Delta \in \{1s, 5s\}$.

103 The deployment rule is deliberately simple. For each workload, engine, and embedding tier, surviving
 104 configurations are ranked by

$$\text{task_cost_est} = C_{\text{retrieval}} + C_{\text{embedding}} + c_{\text{llm_in}} \cdot N_{\text{retrieved_input_tokens}},$$

105 with lower p99 and higher throughput used as tie-breaks. The minimum viable deployment is the
 106 lowest-cost survivor under the default worst-case p99 threshold of 200 ms. The report also exposes
 107 no-p99 cost and quality-first decisions because these are plausible operator objectives with different
 108 winners.

109 2.4 Budget ladder

110 The benchmark uses three budget levels to test whether decisions are stable under shorter local runs.
 111 B0 uses 10 seconds of warmup, 10 seconds of measurement, and one repeat. B1 uses 15 seconds of
 112 warmup, 30 seconds of measurement, and one repeat. B2 uses 30 seconds of warmup, 60 seconds of
 113 measurement, and two repeats. Stability is reported as Spearman rank correlation, top-1 agreement,
 114 and top-2 agreement between budget-level rankings. This design treats instability as a result to report,
 115 not as a failure to hide.

116 3 Experiments

117 3.1 Artifact audit

118 The paper is built from the archived bundle results/20260429T033427Z. The archive con-
 119 tains the run artifacts, generated figures, HOTPOTQA-PORTABLE manifest and checksums, con-
 120 formance matrix, and the operator command documents. A strict-schema validation pass over
 121 artifacts/runs/portable checked 72 run directories and returned zero errors. The paper-facing
 122 support table reports passing conformance and behavior-card coverage for FAISS CPU, LanceDB
 123 in-process, pgvector, and Qdrant; LanceDB service is excluded because it has no conformance row.

124 Table 1 summarizes the strict-latency minimum viable deployments. Under the 200 ms worst-case p99
 125 threshold, all three workloads select FAISS CPU with bge-small. The S1 minimum viable deployment

Table 2: Deployment decisions under strict latency, unconstrained cost, and quality-first objectives. Strict latency uses a 200 ms worst-case p99 threshold.

Workload	Strict p99	No-p99 cost	Quality-first	ρ B0–B2	Stability
S1 single-hop	faiss-cpu / bge-small	faiss-cpu / bge-small	lancedb-inproc / bge-base	0.79	top-1 changed
S2 streaming	faiss-cpu / bge-small	faiss-cpu / bge-small	faiss-cpu / bge-base	0.43	top-1 changed
S3 multi-hop	faiss-cpu / bge-small	lancedb-inproc / bge-small	pgvector / bge-base	0.31	top-1 stable

Table 3: Matched-query S3 quality audit for the quality-first margin. Each row evaluates the same 5,000 HOTPOTQA-PORTABLE queries once with bge-base and `clients_read=4`. Paired deltas are against FAISS HNSW32 in `evidence_coverage@10`; intervals are paired bootstrap 95% intervals.

Engine / search	evidence_coverage@10	p99 (ms)	QPS	Paired Δ evidence_coverage@10
FAISS CPU / HNSW32	0.8712	17.6	247.6	reference
pgvector / IVF32	0.8711	306.0	15.2	−0.0001 [−0.0003, 0.0000]
pgvector / IVF64	0.8711	346.2	14.6	−0.0001 [−0.0003, 0.0000]

reaches nDCG@10 0.506 with a 95% confidence interval of 0.494–0.516 and a maximum observed p99 of 11.2 ms. S2 selects the same engine and embedding, with nDCG@10 0.506, freshness_hit@5s 0.830, and maximum p99 10.0 ms. S3 selects FAISS CPU with bge-small at evidence_coverage@10 0.852 and maximum p99 22.8 ms.

Table 2 shows that the deployment decision is objective-sensitive. The strict p99 and no-p99 cost objectives agree for S1 and S2, but S3 shifts to LanceDB in-process with bge-small when the p99 constraint is removed. Quality-first decisions differ more strongly: LanceDB in-process with bge-base leads S1, FAISS CPU with bge-base leads S2, and pgvector with bge-base leads S3. The S3 quality-first margin is fragile rather than substantive: using all B2 rows, pgvector-base exceeds FAISS-base by only 0.00016 evidence_coverage@10 with an aggregate-row bootstrap interval of −0.00280 to 0.00304.

Table 3 audits that S3 margin with a matched-query check over all 5,000 S3 queries. Under identical bge-base queries, pgvector IVF32 and IVF64 are both 0.0001 evidence_coverage@10 below FAISS HNSW32, with paired intervals that include zero and no positive bootstrap mass at the reported precision. Their p99 latencies are also an order of magnitude larger in this check. We therefore treat the archived pgvector row as objective-sensitivity evidence, not as a meaningful quality advantage.

Table 4 makes the strict-latency decision margins explicit. For S1 and S2, FAISS CPU with bge-small is not a relevance-quality win over the closest LanceDB in-process candidate; it is a cost and quality tie that the deployment rule resolves by lower worst-case p99 and higher throughput. For S3, the nearest strict-cost alternative is not a tie: Qdrant with bge-small is more expensive and has lower evidence coverage, although it is slightly faster at the observed tail. The highest-quality strict-latency S3 option remains FAISS CPU with bge-base, which improves evidence_coverage@10 by 0.020 over bge-small but increases task cost by 0.704 and worst-case p99 by 33.4 ms on the B2 audit rows.

The budget analysis supports a qualified answer to the paper’s research question. B0-to-B2 top-1 agreement is zero for S1 and S2, with Spearman correlations 0.79 and 0.43, respectively. S3 has top-1 agreement of one but Spearman correlation 0.31, which means the recommended minimum deployment is stable while lower-ranked alternatives remain noisy. The benchmark can therefore produce actionable local decisions, but the evidence should include the budget level and stability statistics used to reach those decisions.

3.2 Quick rerun checks

To reduce the risk that the manuscript overfits one archived table, we ran narrow spot checks while preparing the draft. A completed B2 rerun of S1 FAISS CPU with bge-small reproduced the same nDCG@10 value of 0.506 and had maximum p99 27.7 ms across nine measured rows. A completed B2 rerun of S3 FAISS CPU with bge-small reproduced evidence_coverage@10 0.8515 and had maximum p99 20.6 ms across nine measured rows. A completed B2 rerun of S3 FAISS CPU with bge-base produced evidence_coverage@10 0.871 and maximum p99 72.0 ms across nine measured rows, and the full matched-query S3 audit in Table 3 directly checks the quality-first margin. These

Table 4: Strict-decision margins on all B2 rows. The comparison candidate is the next lowest-cost engine/embedding that also meets the workload quality floor and 200 ms worst-case p99 cap. Positive deltas mean the comparison candidate is higher than the strict choice.

Workload	Strict choice	Next strict candidate	Metric	Δ quality	Δ cost	Δ p99	Interpretation
S1 single-hop	faiss-cpu / bge-small	lancedb-inproc / bge-small	nDCG@10	0.000	0.000	+160.9 ms	cost/quality tie; p99 tie-break
S2 streaming	faiss-cpu / bge-small	lancedb-inproc / bge-small	nDCG@10	0.000	0.000	+137.8 ms	cost/quality tie; p99 tie-break
S3 multi-hop	faiss-cpu / bge-small	qdrant / bge-small	evidence_coverage@10	-0.037	+0.548	-4.6 ms	lower cost and higher quality

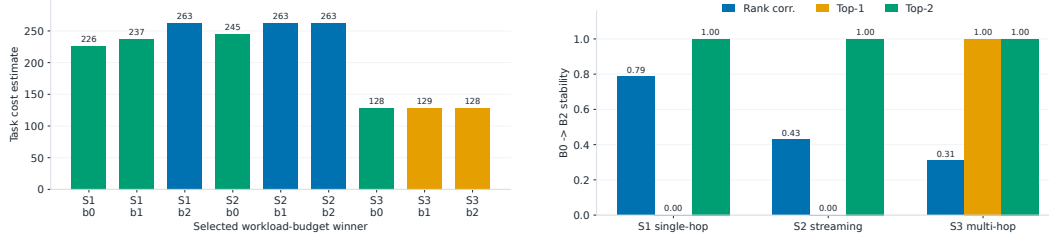


Figure 1: Budget effects in the portable benchmark. Left: task-cost estimates across budget levels. Right: budget-stability summaries show that short runs do not always preserve the B2 ranking.

163 checks are not used to replace the archived paper tables; they are included as reviewer-facing sanity
164 checks for the strict-latency and objective-sensitivity story.

165 Table 5 adds a larger same-machine S2 competitor check. Under one generated matrix, FAISS CPU
166 and Qdrant each complete two B2 repeats with zero errors, 894 quality observations per repeat, and
167 100 freshness probes per repeat. Qdrant matches FAISS exactly on all 200 paired freshness events
168 and has a paired nDCG@10 delta of -0.0027 with a 95% interval of -0.0052 to -0.0006 over 1,788
169 matched quality observations. This check rules out a hidden Qdrant quality or freshness win in this
170 setting. The p99 direction differs from the earlier one-row Qdrant check, so we treat S2 latency as
171 local-run evidence and keep the archived deployment table as the primary decision source.

172 The earlier same-orchestration deadline mini-bundle is retained as a turnaround-oriented repeatability
173 check. It used smaller static and freshness caps; both engines completed with zero errors, Qdrant
174 minus FAISS paired nDCG@10 was -0.0007 with a 95% interval of -0.0026 to 0.0005 over 500
175 matched quality observations, and the paired freshness deltas were exactly zero over 80 matched
176 freshness events.

177 4 Limitations and broader impact

178 MaxionBench-Portable is a bounded benchmark, not a claim that commodity hardware can substitute
179 for production-scale evaluation. The experiments run on one local operator machine, so the results
180 do not characterize distributed retrieval, GPU indexing, cloud variability, or multi-tenant production
181 workloads. S3 uses HOTPOTQA-PORTABLE, a frozen paragraph-level proxy for full-Wikipedia
182 multi-hop retrieval, and S2 measures top-k retrievability after inserts rather than lower-level index
183 visibility. These choices make the benchmark reproducible and one-day executable, but they narrow
184 the scope of the claims.

185 The uncertainty estimates are conditional on this fixed benchmark setting. Confidence intervals are
186 query- or event-level summaries, not hardware-population intervals, and observed p99 latency is not
187 a production tail-latency guarantee. The current result bundle also shows why short benchmark runs
188 should not be overinterpreted. S1 and S2 change top-1 recommendations between B0 and B2, so
189 a screening run is not enough to make a final deployment decision for those workloads. S3 keeps
190 the same top-1 recommendation but has low full-rank correlation, indicating that the best decision
191 can be stable even when the rest of the ranking is not. Future versions should add more repeated B2
192 runs, more event streams, dedicated S2 competitor reruns, a second hardware profile, and an explicit
193 production-sized stress track.

194 The broader impact of MaxionBench-Portable is primarily methodological. A reproducible benchmark
195 can help operators choose lower-cost, lower-latency retrieval infrastructure without hiding quality or
196 freshness tradeoffs. The negative risk is that readers may treat the benchmark winner as a universal

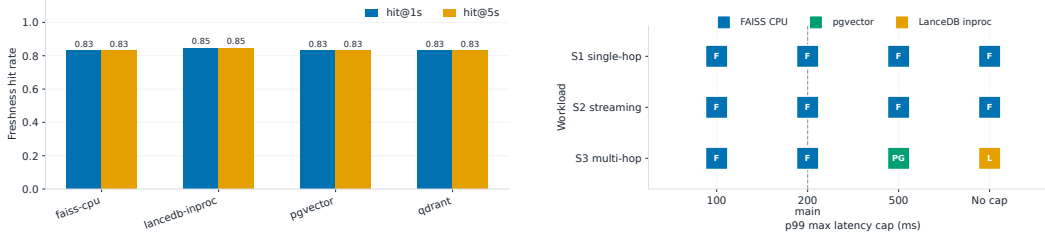


Figure 2: Agent-facing stressors. Left: S2 freshness is measured after acknowledged inserts at fixed one- and five-second probes. Right: S3 is sensitive to relaxing the p99 threshold, while S1 and S2 keep the same minimum viable engine across the reported thresholds.

Table 5: S2 larger same-machine competitor check with bge-small, clients_read=8, clients_write=2, two repeats per engine, 1,788 matched quality queries, and 200 matched freshness events. Paired deltas are Qdrant minus FAISS with 95% paired bootstrap intervals.

Engine / search	nDCG@10	Paired Δ nDCG@10	freshness_hit@5s	Mean / max p99 (ms)	Errors
FAISS CPU / ef64	0.5055	reference	0.840	47.8 / 74.1	0
Qdrant / ef64	0.5028	-0.0027 [-0.0052, -0.0006]	0.840	17.1 / 20.7	0

recommendation rather than a result under a specific machine, dataset, objective, and conformance policy. The manuscript mitigates that risk by reporting exclusions, limitations, budget stability, and objective sensitivity directly.

5 Conclusion

MaxionBench-Portable provides a conformance-gated, one-day local protocol for evaluating retrieval infrastructure for agentic AI. The archived results show that strict-latency deployment choices can be made reproducibly on commodity hardware, but they also show that quality-first and unconstrained-cost objectives can select different engines and that some strict decisions are latency tie-breaks rather than quality wins. The main lesson is not that one engine dominates; it is that deployment benchmarks for agentic retrieval should report quality, freshness, p99 latency, task cost, conformance, budget stability, and decision margins together. The next step is to turn this portable audit into a broader stress track with longer streaming-memory competitor runs and multiple hardware profiles.

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Table 6: Reviewer-facing evidence map. The main claims are framed as benchmark and decision-audit claims rather than universal engine rankings.

Claim	Main evidence	Residual risk	Paper stance
Conformance-gated reporting	72 strict-schema run dirs; support table	Adapter coverage is local	Supported protocol claim
Strict-latency decisions	Main and decision tables; margin table	One hardware profile	Local deployment evidence
S2 has no hidden Qdrant quality/freshness win	Larger same-machine paired S2 check in Table 5 plus the bounded mini-bundle	Freshness remains capped	Sanity check, not universal latency evidence
S3 pvector quality edge is not substantive	5,000-query paired audit in Table 3	One matched audit	Objective sensitivity, not quality win
Budget stability is workload-dependent	B0-to-B2 top-1 and Spearman stats	Lower ranks remain noisy	Report budgets explicitly

Nandan Thakur, Nils Reimers, Andreas Rücklé, Abhishek Srivastava, and Iryna Gurevych. BEIR: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. In *Advances in Neural Information Processing Systems*, 2021.

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Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018.

A Reproducibility details

The portable workflow is run from the repository root. The main commands are:

```
maxionbench portable-workflow setup --json
maxionbench portable-workflow data --json
maxionbench submit-portable --budget b0 --json
maxionbench submit-portable --budget b1 --json
maxionbench submit-portable --budget b2 --json
maxionbench portable-workflow finalize --json
```

The archived bundle used by this manuscript is results/20260429T033427Z. It contains 972 run-artifact files, 42 generated figure files, 27 HOTPOTQA-PORTABLE files, and 27 conformance files. The strict-schema validation command used during manuscript preparation was:

```
python -m maxionbench.cli validate --input artifacts/runs/portable \
--strict-schema --json
```

It returned "pass": true, "error_count": 0, and "run_dirs_checked": 72.

The archived portable matrix contains 72 results.parquet files, including 24 B2 result files with matching run_metadata.json files. The recorded local runtime is an Apple M4 Mac mini with 10 logical CPU cores, 16 GB RAM, macOS 26.4.1, Python 3.11.14, Docker 29.4.0, and no GPUs. HOTPOTQA-PORTABLE contains 66,635 documents, 7,405 questions, and 14,810 qrels; hotpot_portable/checksums.json stores SHA-256 checksums for the corpus, queries, qrels, metadata, and manifest.

The reviewer artifact also includes an artifact card, a local readiness verifier, and Croissant-style metadata for HOTPOTQA-PORTABLE. The verifier checks that the manuscript, archive manifest, paper-facing tables, paired-audit summaries, and metadata files are present and internally consistent; the Croissant metadata records the corpus, query, and qrel files, their intended evaluation use, and the license/provenance boundary inherited from HotpotQA.

B Claim-evidence map

- Claim: MaxionBench-Portable is conformance-gated. Evidence: the support table reports passing conformance and behavior cards for all included engines, and the report excludes LanceDB service because conformance is missing.

- Claim: strict 200 ms p99 deployment selects FAISS CPU with bge-small for S1, S2, and S3. Evidence: Table 1 and Table 2.
- Claim: strict decisions have different margins across workloads. Evidence: Table 4 shows S1/S2 as cost/quality ties resolved by latency and S3 as lower-cost and higher-quality than the next strict-cost candidate.
- Claim: objective choice changes deployment conclusions. Evidence: Table 2 shows different quality-first winners for all workloads and a no-p99 cost shift for S3.
- Claim: the S3 quality-first pgvector result should not be read as a substantive quality advantage. Evidence: Table 3 reports a matched-query audit over all 5,000 S3 queries in which pgvector trails FAISS CPU by 0.0001 evidence_coverage@10 with paired 95% interval [-0.0003, 0.0000].
- Claim: the S2 strict decision is not hiding a Qdrant quality or freshness win. Evidence: Table 5 reports a larger same-machine paired check with 1,788 matched quality observations and 200 matched freshness events.
- Claim: budget stability is workload-dependent. Evidence: B0-to-B2 top-1 agreement is 0 for S1/S2 and 1 for S3, with Spearman correlations 0.79, 0.43, and 0.31.

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes].

Justification: The abstract and introduction claim a conformance-gated local benchmark and explicitly limit the stability claim; Tables 1 and 2 provide the supporting evidence.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes].

Justification: The manuscript includes a dedicated limitations section describing the single-node setting, bounded S3 corpus, S2 freshness interpretation, and budget-stability limits.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete proof?

Answer: [N/A].

Justification: The paper does not present theoretical results; it defines benchmark metrics and decision rules.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes].

Justification: The appendix and artifact README list the workflow commands, archive identifier, validation command, runtime metadata, checksums, report artifacts, and local artifact-readiness verifier.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?

Answer: [Yes].

Justification: The benchmark code, commands, generated manifests, and archived result bundle are part of the repository prepared with this manuscript; any public release should preserve anonymization at submission time.

6. Experimental setting/details

311 Question: Does the paper specify all training and test details necessary to understand the
 312 results?

313 Answer: [Yes].

314 Justification: The benchmark-design section specifies engines, datasets, embedding tiers,
 315 concurrency pins, budget levels, metrics, decision rules, and conformance gates.

316 **7. Experiment statistical significance**

317 Question: Does the paper report error bars suitably and correctly defined or other appropriate
 318 information about statistical significance?

319 Answer: [Yes].

320 Justification: The main table reports 95% confidence intervals; targeted S2/S3 audits report
 321 paired bootstrap intervals for the highest-risk margins.

322 **8. Experiments compute resources**

323 Question: For each experiment, does the paper provide sufficient information on compute
 324 resources needed to reproduce the experiments?

325 Answer: [Yes].

326 Justification: The scope and reproducibility sections state the single-node commodity-
 327 hardware setting and exact budget ladder; the archived run metadata records per-run resource
 328 fields.

329 **9. Code of ethics**

330 Question: Does the research conducted in the paper conform with the NeurIPS Code of
 331 Ethics?

332 Answer: [Yes].

333 Justification: The work evaluates retrieval infrastructure using public benchmark artifacts
 334 and does not involve human subjects, private data, or model training on sensitive data.

335 **10. Broader impacts**

336 Question: Does the paper discuss both potential positive societal impacts and negative
 337 societal impacts of the work performed?

338 Answer: [Yes].

339 Justification: The limitations and broader-impact section discusses the benefit of transparent
 340 deployment evidence and the risk of overgeneralizing local benchmark winners.

341 **11. Safeguards**

342 Question: Does the paper describe safeguards for responsible release of data or models that
 343 have a high risk for misuse?

344 Answer: [N/A].

345 Justification: The paper does not release new generative models or scraped high-risk datasets;
 346 it uses existing retrieval benchmark assets.

347 **12. Licenses for existing assets**

348 Question: Are creators or original owners of assets properly credited and are licenses and
 349 terms of use explicitly mentioned and respected?

350 Answer: [Yes].

351 Justification: The manuscript cites the datasets, software systems, and embedding models
 352 used; final submission should include any repository-level license manifests alongside the
 353 artifacts.

354 **13. New assets**

355 Question: Are new assets introduced in the paper well documented and is the documentation
 356 provided alongside the assets?

357 Answer: [Yes].

358 Justification: HOTPOTQA-PORTABLE is documented through its preprocessing description,
 359 manifest, checksums, archived bundle, and Croissant-style metadata included with the
 360 reviewer artifact.

361 **14. Crowdsourcing and research with human subjects**

362 Question: For crowdsourcing experiments and research with human subjects, does the paper
 363 include instructions, screenshots, and compensation details?

364 Answer: [N/A].

365 Justification: The paper does not involve crowdsourcing or human-subject experiments.

366 **15. Institutional review board approvals or equivalent for research with human subjects**

367 Question: Does the paper describe risks to study participants and IRB approvals or equivalent?
 368

369 Answer: [N/A].

370 Justification: The paper does not involve human-subject research.

371 **16. Declaration of LLM usage**

372 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 373 non-standard component of the core methods?

374 Answer: [N/A].

375 Justification: LLMs are not an important, original, or non-standard component of the
 376 benchmark method; the benchmark evaluates retrieval infrastructure and local embedding
 377 models.